

Table 1. Full MGSM multilingual-transfer results for Qwen3-4B, Llama-3.2-3B-Instruct, Phi-4-mini-instruct, and Granite-3.3-2B-Instruct on 100 matched questions per language. Values are full-sample means $\pm$ ten-group matched-question jackknife standard errors. Higher is better for every column.

Model	Language	Method	Accuracy (%) $\uparrow$	Spanish relevance (0–2) $\uparrow$	Instruction relevance (0–2) $\uparrow$	Fluency (0–2) $\uparrow$
Qwen3-4B	Chinese	Original	71.0 $\pm$ 4.1	0.00 $\pm$ 0.00	1.80 $\pm$ 0.05	1.71 $\pm$ 0.05
		A-LQR	20.0 $\pm$ 2.6	1.98 $\pm$ 0.02	1.53 $\pm$ 0.03	1.15 $\pm$ 0.03
		H_∞ (ours)	51.0 $\pm$ 3.5	2.00 $\pm$ 0.00	1.87 $\pm$ 0.05	1.52 $\pm$ 0.04
	French	Original	60.0 $\pm$ 4.5	0.00 $\pm$ 0.00	1.93 $\pm$ 0.03	1.70 $\pm$ 0.04
		A-LQR	29.0 $\pm$ 5.0	1.94 $\pm$ 0.04	1.62 $\pm$ 0.05	1.11 $\pm$ 0.03
		H_∞ (ours)	48.0 $\pm$ 4.7	1.96 $\pm$ 0.04	1.96 $\pm$ 0.02	1.42 $\pm$ 0.07
	Japanese	Original	56.0 $\pm$ 4.8	0.00 $\pm$ 0.00	1.80 $\pm$ 0.04	1.64 $\pm$ 0.04
		A-LQR	32.0 $\pm$ 2.5	1.84 $\pm$ 0.06	1.63 $\pm$ 0.06	1.01 $\pm$ 0.03
		H_∞ (ours)	40.0 $\pm$ 4.9	1.98 $\pm$ 0.02	1.71 $\pm$ 0.07	1.20 $\pm$ 0.05
	Swahili	Original	15.0 $\pm$ 4.3	0.00 $\pm$ 0.00	1.43 $\pm$ 0.07	1.23 $\pm$ 0.05
		A-LQR	12.0 $\pm$ 3.6	<u>1.40 $\pm$ 0.09</u>	1.32 $\pm$ 0.07	0.97 $\pm$ 0.03
		H_∞ (ours)	14.0 $\pm$ 3.7	0.92 $\pm$ 0.12	1.44 $\pm$ 0.07	1.06 $\pm$ 0.04
	Telugu	Original	6.0 $\pm$ 2.7	0.00 $\pm$ 0.00	1.83 $\pm$ 0.04	1.05 $\pm$ 0.02
		A-LQR	<u>19.0 $\pm$ 2.8</u>	<u>1.88 $\pm$ 0.04</u>	<u>1.46 $\pm$ 0.06</u>	<u>1.18 $\pm$ 0.03</u>
		H_∞ (ours)	<u>8.0 $\pm$ 2.5</u>	0.34 $\pm$ 0.08	1.33 $\pm$ 0.04	1.05 $\pm$ 0.03
Llama-3.2-3B	Chinese	Original	43.0 $\pm$ 3.3	0.00 $\pm$ 0.00	1.79 $\pm$ 0.07	1.59 $\pm$ 0.07
		A-LQR	<u>26.0 $\pm$ 3.7</u>	1.80 $\pm$ 0.04	<u>1.49 $\pm$ 0.08</u>	1.21 $\pm$ 0.08
		H_∞ (ours)	<u>22.0 $\pm$ 4.7</u>	1.98 $\pm$ 0.02	1.46 $\pm$ 0.06	1.35 $\pm$ 0.05
	French	Original	49.0 $\pm$ 4.6	0.00 $\pm$ 0.00	1.83 $\pm$ 0.03	1.71 $\pm$ 0.04
		A-LQR	<u>43.0 $\pm$ 4.2</u>	<u>1.24 $\pm$ 0.09</u>	<u>1.76 $\pm$ 0.04</u>	<u>1.09 $\pm$ 0.03</u>
		H_∞ (ours)	<u>6.0 $\pm$ 2.7</u>	0.66 $\pm$ 0.06	0.53 $\pm$ 0.04	0.47 $\pm$ 0.04
	Japanese	Original	27.0 $\pm$ 3.0	0.00 $\pm$ 0.00	1.56 $\pm$ 0.08	1.34 $\pm$ 0.09
		A-LQR	<u>26.0 $\pm$ 3.7</u>	1.94 $\pm$ 0.04	<u>1.68 $\pm$ 0.08</u>	<u>1.31 $\pm$ 0.07</u>
		H_∞ (ours)	<u>12.0 $\pm$ 3.3</u>	1.98 $\pm$ 0.02	1.31 $\pm$ 0.10	1.16 $\pm$ 0.08
	Swahili	Original	40.0 $\pm$ 3.3	0.00 $\pm$ 0.00	1.79 $\pm$ 0.02	1.49 $\pm$ 0.03
		A-LQR	<u>22.0 $\pm$ 4.4</u>	0.68 $\pm$ 0.10	<u>1.47 $\pm$ 0.04</u>	0.98 $\pm$ 0.04
		H_∞ (ours)	<u>14.0 $\pm$ 4.0</u>	1.90 $\pm$ 0.04	1.28 $\pm$ 0.06	1.05 $\pm$ 0.05
	Telugu	Original	6.0 $\pm$ 2.2	0.00 $\pm$ 0.00	1.43 $\pm$ 0.04	1.07 $\pm$ 0.03
		A-LQR	<u>21.0 $\pm$ 6.0</u>	<u>1.86 $\pm$ 0.05</u>	<u>1.44 $\pm$ 0.08</u>	<u>1.25 $\pm$ 0.07</u>
		H_∞ (ours)	<u>7.0 $\pm$ 2.6</u>	0.82 $\pm$ 0.06	0.53 $\pm$ 0.10	0.54 $\pm$ 0.09

Table 1 (continued). Full MGSM multilingual-transfer results.

Model	Language	Method	Accuracy (%) $\uparrow$	Spanish relevance (0–2) $\uparrow$	Instruction relevance (0–2) $\uparrow$	Fluency (0–2) $\uparrow$
Phi-4-mini	Chinese	Original	63.0 $\pm$ 5.2	0.00 $\pm$ 0.00	1.90 $\pm$ 0.04	1.80 $\pm$ 0.04
		A-LQR	0.0 $\pm$ 0.0	1.24 $\pm$ 0.08	0.07 $\pm$ 0.02	0.10 $\pm$ 0.03
		H_∞ (ours)	47.0 $\pm$ 4.2	1.64 $\pm$ 0.09	1.72 $\pm$ 0.05	1.39 $\pm$ 0.07
	French	Original	58.0 $\pm$ 3.6	0.00 $\pm$ 0.00	1.91 $\pm$ 0.02	1.64 $\pm$ 0.05
		A-LQR	0.0 $\pm$ 0.0	0.70 $\pm$ 0.07	0.11 $\pm$ 0.05	0.07 $\pm$ 0.03
		H_∞ (ours)	49.0 $\pm$ 5.0	1.40 $\pm$ 0.05	1.90 $\pm$ 0.03	0.98 $\pm$ 0.03
	Japanese	Original	49.0 $\pm$ 3.8	0.00 $\pm$ 0.00	1.91 $\pm$ 0.03	1.57 $\pm$ 0.04
		A-LQR	0.0 $\pm$ 0.0	0.94 $\pm$ 0.09	0.00 $\pm$ 0.00	0.03 $\pm$ 0.02
		H_∞ (ours)	41.0 $\pm$ 4.1	0.62 $\pm$ 0.10	1.68 $\pm$ 0.06	1.20 $\pm$ 0.05
	Swahili	Original	32.0 $\pm$ 3.9	0.00 $\pm$ 0.00	1.67 $\pm$ 0.05	1.36 $\pm$ 0.08
		A-LQR	0.0 $\pm$ 0.0	0.92 $\pm$ 0.10	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
		H_∞ (ours)	21.0 $\pm$ 2.8	1.00 $\pm$ 0.14	1.18 $\pm$ 0.09	0.77 $\pm$ 0.08
	Telugu	Original	25.0 $\pm$ 3.7	0.00 $\pm$ 0.00	1.57 $\pm$ 0.04	1.25 $\pm$ 0.05
		A-LQR	0.0 $\pm$ 0.0	1.64 $\pm$ 0.10	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
		H_∞ (ours)	18.0 $\pm$ 2.0	1.38 $\pm$ 0.10	1.34 $\pm$ 0.04	0.89 $\pm$ 0.07
Granite-3.3-2B	Chinese	Original	37.0 $\pm$ 5.0	0.00 $\pm$ 0.00	1.67 $\pm$ 0.05	1.50 $\pm$ 0.06
		A-LQR	30.0 $\pm$ 3.0	0.06 $\pm$ 0.03	1.59 $\pm$ 0.05	1.12 $\pm$ 0.04
		H_∞ (ours)	39.0 $\pm$ 3.1	0.00 $\pm$ 0.00	1.70 $\pm$ 0.04	1.53 $\pm$ 0.04
	French	Original	33.0 $\pm$ 4.5	0.00 $\pm$ 0.00	1.68 $\pm$ 0.06	1.43 $\pm$ 0.04
		A-LQR	35.0 $\pm$ 4.8	0.02 $\pm$ 0.02	1.53 $\pm$ 0.04	1.31 $\pm$ 0.03
		H_∞ (ours)	28.0 $\pm$ 4.4	0.00 $\pm$ 0.00	1.71 $\pm$ 0.07	1.39 $\pm$ 0.04
	Japanese	Original	29.0 $\pm$ 3.8	0.00 $\pm$ 0.00	1.60 $\pm$ 0.05	1.33 $\pm$ 0.06
		A-LQR	23.0 $\pm$ 4.2	0.00 $\pm$ 0.00	1.48 $\pm$ 0.06	1.19 $\pm$ 0.06
		H_∞ (ours)	28.0 $\pm$ 3.6	0.00 $\pm$ 0.00	1.67 $\pm$ 0.04	1.38 $\pm$ 0.04
	Swahili	Original	6.0 $\pm$ 2.2	0.00 $\pm$ 0.00	0.69 $\pm$ 0.05	0.64 $\pm$ 0.07
		A-LQR	2.0 $\pm$ 1.3	0.04 $\pm$ 0.03	0.32 $\pm$ 0.08	0.40 $\pm$ 0.07
		H_∞ (ours)	6.0 $\pm$ 3.4	0.00 $\pm$ 0.00	0.48 $\pm$ 0.04	0.60 $\pm$ 0.05
	Telugu	Original	6.0 $\pm$ 2.7	0.00 $\pm$ 0.00	1.07 $\pm$ 0.05	1.01 $\pm$ 0.04
		A-LQR	4.0 $\pm$ 2.2	0.06 $\pm$ 0.03	0.28 $\pm$ 0.04	0.00 $\pm$ 0.00
		H_∞ (ours)	5.0 $\pm$ 2.2	0.00 $\pm$ 0.00	1.08 $\pm$ 0.04	0.98 $\pm$ 0.04